

Experiment#05

Objective:

To make 2x1 multiplexer using basic logic gate and 2x4 line decoder using basic logic gate.

Equipment's:

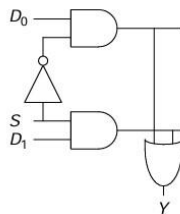
- Breadboard
- Jumper wires
- Power supply
- Integrated ICs
- LED light

Theory:

Multiplexers

Multiplexers are among the most commonly used combinational circuits. They choose an output from among several possible inputs based on the value of a select signal. A multiplexer is sometimes affectionately called a mux.

the schematic and truth table for a 2:1 multiplexer with two data inputs, D_0 and D_1 , a select input, S , and one output, Y . The multiplexer chooses between the two data inputs based on the select: if $S = 0$, $Y = D_0$, and if $S = 1$, $Y = D_1$. S is also called a control signal because it controls what the multiplexer does.

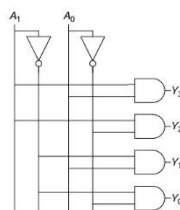


Fig#01

Decoder:

A decoder has N inputs and 2^N outputs. It asserts exactly one of its outputs depending on the input combination. When $A_1A_0 = 00$, Y_0 is 1. When $A_1A_0 = 01$, Y_1 is 1. And so forth. The outputs are called one-hot, because exactly one is "hot" (HIGH) at a given time.

For the 2:4 decoder using four AND gates. Each gate depends on either the true or the complementary form of each input. In general, an $N:2^N$ decoder can be constructed from 2^N N -input AND gates that accept the various combinations of true or complementary inputs. Each A_1A_0 output in a decoder represents a single minterm. For example, Y_0 represents the minterm $\overline{A_1}\overline{A_0}$. This fact will be handy when using decoders with other digital building blocks.

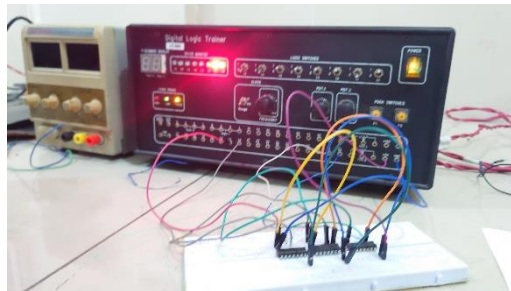


Fig#02

Procedure:

- First, collect all the required equipment and components that are used.
- Now make a circuit diagram according to the given equation.
- Make a circuit on the breadboard according to the circuit diagram.
- Put LED/DMM to check our circuit
- After making the circuit give a power supply to the circuit with the help of DC power supply and jumper wires.
- Check all the connections have been connected according circuit diagram.
- Now give supply to the circuit to make the truth table by changing values.

Experimental Setup:



Fig#03

Truth table:

Multiplexer:

[illegible]

Decoder:

[illegible]

Conclusion:

In this experiment, we have learned about multiplexers. Its circuit diagram, making, working, and how to make a truth table with the readings that were calculated during the experiment. We also learn about the decoder its circuit diagram, making, working, and truth table.